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An LLM-Based Conversational AI Dispatcher for Automating Taxi Phone Orders in Bulgaria

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Abstract –**This paper presents an architecture for automating Bulgarian taxi phone orders. A low-latency real-time LLM core handles voice processing and utilizes dynamic function calling to invoke external APIs for critical pick-up address verification and dispatch integration. SIP telephony and in-process .NET system integration demonstrates validated robust operational performance.**

Keywords – **Conversational Real-time AI; LLM; STT; TTS; PABX.**

I. Introduction

Modern urban mobility continues to face substantial challenges due to the coexistence of legacy telephony systems and contemporary digital applications. In Bulgaria, longstanding regulatory mandates have resulted in the persistent use of traditional phone-based taxi orders. For example, Triumph Taxi in Varna processes approximately 10,000 calls daily [1], leading to elevated operational costs, delayed response times during peak hours, and considerable dispatcher workload.

An automated taxi dispatcher is presented, integrating advanced large language models (LLMs) with robust telephony infrastructure. The system utilizes dynamic function calling mechanisms to interpret and execute customer requests in real time, while maintaining compatibility with existing dispatch infrastructure. In addition to reducing manual intervention, the approach addresses the need for service continuity in environments where regulatory barriers impede digital transformation. The approach also enables incremental automation and further improvement through adaptive learning and user feedback.

II. Background and Problem Justification

Operational Bottleneck

In contrast to global trends favoring app-based services, Bulgaria's taxi industry continues to process a substantial proportion of orders via traditional phone calls. Approximately 40% of orders are processed manually, increasing labor requirements and resulting in slower service delivery. The high call volume - around 10,000 daily at Triumph Taxi [1] - emphasizes the inefficiencies of legacy systems and highlights the need for automation to reduce dispatcher workload and operational errors.

Regulatory Constraints and Legacy Infrastructure

Bulgarian regulations, such as the ordinances from 1999 mandating specific hardware fiscal devices [2], limit the adoption of fully integrated app-based ordering systems. Such legal requirements necessitate that modernization efforts focus on enhancing the existing phone channel rather than replacing it entirely. The proposed system is developed in accordance with these constraints, improving efficiency without requiring a complete overhaul of established infrastructure.

III. Related Work

The automation of task-oriented dialogues, particularly in voice-based call centers, has led to extensive research in conversational AI [3]. Early systems divided the process into separate modules, including speech recognition (ASR/STT), natural language understanding (NLU), dialogue state tracking (DST), dialogue policy optimization, and natural language generation (NLG) combined with text-to-speech (TTS) [4]. Although these systems proved effective in constrained domains, they frequently struggled to maintain conversational fluidity and context.

Recent advances in pre-trained LLMs provide more cohesive, end-to-end conversational capabilities [5]. Studies indicate that models such as GPT-4 [6] excel at few-shot learning and can handle complex dialogue flows with minimal task-specific training. Brown et al. (2020) [7] highlighted the strengths of LLMs in understanding nuanced user intents. Furthermore, the introduction of function calling, as described by Nakano et al. (2021) [8] and Schick et al. (2023) [9], enables these systems to interact dynamically with external databases and APIs, supporting real-time actions beyond static response generation.

These advancements have enabled the integration of a specialized LLM core with dynamic function calling [8, 9, 10] within a telephony environment. While most existing research focuses on high-resource languages and text-based interactions, this approach is specifically adapted for the Bulgarian language, with particular emphasis on live call processing in taxi dispatch scenarios. This adaptation marks a significant step beyond generic LLM applications toward real-time operational deployment. Unlike most prior work, this system addresses live telephony-based taxi dispatch in Bulgarian, a low-resource and highly regulated context, and is validated in operational conditions with real users.

IV. Analysis of Phone Order Structure

Comprehensive understanding of the phone order process is essential in designing an effective automated system. Standard taxi call procedures involve the following key data exchanges:

Input Information

- Origin Address (pick-up location)

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- Destination Address (optional but often provided)
- Pick-up Time (immediate or scheduled)

Output Information (to the Customer)

- Dispatch Confirmation
- Assigned Vehicle Identifier (taxi number)
- Estimated Time of Arrival (ETA)

Although digital channels (web, app, social media) handle approximately 60% of taxi orders, the remaining 40% processed via traditional phone calls continue to require significant resources, highlighting the need for automation specifically tailored to the conventional phone channel.

Examination of call structure reveals that a typical successful call comprises several distinct phases:

- Greeting and initial identification (approximately 1–5 seconds).
- Obtaining and capturing of the origin address (approximately 5 seconds, with potential clarifications extending this duration).
- Optional capture of the destination address (similarly timed).
- A system query phase for vehicle search (ranging from 10 to 60 seconds).
- Final confirmation or failure notification (approximately 5 seconds).

Each phase introduces potential delays due to manual processing. The proposed automation system is designed to significantly reduce the cumulative time spent during these interactions, thereby optimizing the overall call handling process.

V. Proposed AI Dispatcher Architecture

The proposed architecture consists of four integrated components:

- **PABX Integration**: An open-source PABX (e.g., Asterisk) designed to manage high-volume call routing, bidirectional STT/TTS audio streaming, and LLM-instructed call control and rerouting.
- **Speech Processing**: Real-time STT (Bulgarian, handling dialects/noise) transcribes speech; concurrent Bulgarian-optimized TTS synthesizes responses.
- **LLM Core with Dynamic Function Calling**: A custom LLM, optimized for rapid inference and taxi-specific dialogue, orchestrates tasks using dynamic function calls.
- **External API Interfaces**: Enable real-time geocoding, address validation, and reliable data exchange with legacy dispatch systems.

Telephony and PABX Integration

At the front end, an open-source PABX system (such as Asterisk) receives incoming calls and directs audio streams to the STT engine. This module is designed to handle high call volumes with minimal latency and incorporates mechanisms to reroute calls to human operators if automated processing encounters difficulties. This redundancy maintains service quality even under challenging conditions.

Speech Processing Module (STT/TTS)

A state-of-the-art STT engine is implemented to accurately transcribe spoken Bulgarian, effectively handling background noise and varied accents. In parallel, a TTS engine, optimized for the nuances of Bulgarian speech, generates context-appropriate audio responses. These modules employ advanced signal processing techniques, including noise suppression, acoustic model adaptation, and accent normalization, to reliably capture and reproduce speech in real time.

Real-Time LLM Core with Dynamic Function Calling

The central component of the architecture is a dedicated real-time LLM core specifically designed for taxi dispatch applications. This core has several key attributes:

- Ultra Low-Latency Inference: Optimized hardware and neural architecture enable live call inference in tens of milliseconds.
- Domain-Specific Training: Adapted using extensive anonymized data (approximately one year of taxi dialogues, millions in total) and simulated scenarios. Preprocessing (normalization, augmentation) addresses Bulgarian dialects, colloquialisms, address formats, and common requests, ensuring robust interpretation.
- Dynamic Function Calling: Advanced routines enable instant external API invocation [11] by the LLM core. For instance, if a caller provides an ambiguous address, the system automatically triggers an address validation process and, if confirmed, initiates a taxi dispatch command - all within a single conversational turn.

External API Interfaces

Robust API connections are established to interact with modern geocoding services and legacy taxi dispatch systems. These APIs validate and normalize address data and automatically allocate vehicles based on real-time availability. They are designed with built-in error correction and retry logic to ensure reliability, even in the presence of temporary network disruptions.

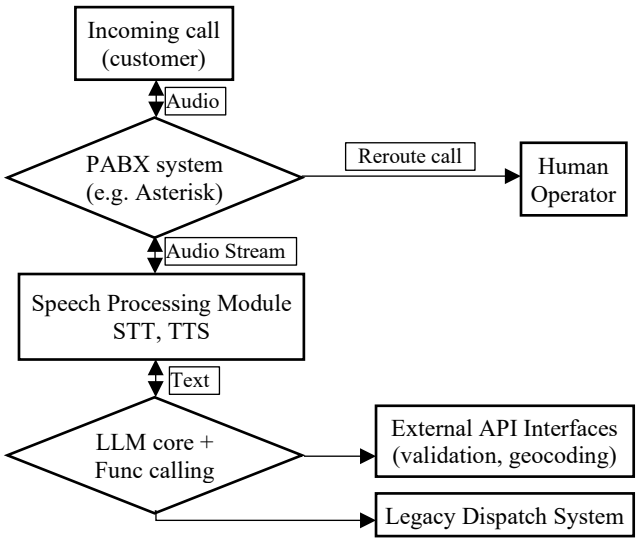

Fig. 1. Architecture of the Conversational AI Dispatcher.

The architecture (see Fig. 1) integrates SIP telephony, real-time STT and TTS modules, and a domain-adapted LLM core with dynamic function calling. The LLM orchestrates dialogue and API integration, while seamless fallback to human operators is ensured.

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VI. Experimental Validation

Pilot tests were carried out in collaboration with Triumph Taxi in Varna [1]. Validation involved simulations of both standard and ambiguous request scenarios. Performance metrics were collected across multiple test runs; key indicators are summarized in Table I.

Validation Methodology

The experimental procedure involved the following steps:

- **Test Data Preparation**: A comprehensive suite of simulated call scenarios was developed, incorporating both standard and complex request patterns. These included ambiguous and noisy addresses, regional dialectal variations, and real-world conversational interruptions, to reflect the operational complexity encountered in practice.
- **System Configuration**: Deployment and calibration of the complete system stack-including PABX integration, STT/TTS modules, and the real-time LLM core with dynamic function calling - for the operational environment.
- **Simulated Call Execution**: Systematic execution and recording of calls, with audio and processing data logged for detailed analysis.
- **Data Collection and Analysis**: Measurement of key performance indicators (KPIs), including address recognition accuracy, task completion rates, turn response latency, and average call duration. These metrics were benchmarked against performance baselines established by human-operated dispatch systems.

Table I
EXPERIMENTAL TEST RESULTS

Metric	Unit	Method	Result	Baseline
Accuracy - Address Recognition	%	%correctly identified addresses	92%	~98%
Task Completion Rate	%	% success processed	85%	~95%
Interaction quality	score (1-5)	Subjective rating	3.8	~4.5
Latency - Turn Response	sec.	Average response time	0.7s	~0.5 - 1
Avg. Call Duration	sec.	Avg. call duration	~50s	~40 - 90

Key Performance Indicators and Results

The primary performance metrics are as follows:

- **Address Recognition Accuracy**: 92%-reflecting the percentage of correctly identified and validated addresses.
- **Task Completion Rate**: 85%-indicating the proportion of calls completed end-to-end without human intervention.
- **Interaction quality** - Reflects user experience and conversation smoothness.
- **Average Turn Response Latency**: Approximately 0.7 seconds-measuring the time from the end of user speech to the system's response.
- **Average Call Duration**: Approximately 50 seconds-a reduction from the 40–90 seconds typically observed in manual processing.

Qualitative feedback from dispatch operators and pilot users indicated improvements in consistency, response times, and conversational flow. These findings demonstrate the benefits of integrating a dedicated real-time LLM core with dynamic function calling, which together enhance operational speed and accuracy while reducing overall dispatcher workload.

User feedback rated interaction quality as satisfactory, supporting the system's potential for reducing manual workload and improving operational efficiency.

VII. Expected Benefits and Technical Challenges

Deployment of the automated dispatcher offers several significant advantages:

- **Enhanced Operational Efficiency**: Reductions in call handling time and decreased manual intervention directly lower operational costs.
- **Reduced Dispatcher Workload**: Routine and standard queries are processed automatically, enabling human operators to focus on complex or emergency cases.
- **Improved Service Consistency**: Automated, standardized responses ensure reliable and predictable customer interactions.
- **Scalability**: The architecture seamlessly adapts to surges in call volume during peak periods without necessitating proportional increases in staffing.
- **Data-Driven Insights**: System logs and interaction analytics enable continuous performance refinement and strategic decision-making.
- **Regulatory Compliance**: The system is engineered to operate within existing telephony and hardware frameworks, conforming to stringent regulatory standards while leveraging modern AI innovations.

However, several challenges remain to fully realize the system's potential:

- **Optimizing STT/TTS for Bulgarian**: Although current models perform well, ongoing tuning is necessary to accommodate diverse dialects, rapid speech, and background noise effectively.
- **Enhancing Robustness of the LLM Core**: Further fine-tuning is required to better manage ambiguous phrases, minimize inaccuracies, and retain contextual information over prolonged interactions.
- **Ensuring Reliable API Integration**: Establishing and maintaining fault-tolerant connections with both modern geocoding services and legacy dispatch systems remains a complex engineering task.
- **Developing Comprehensive Error Handling Protocols**: Robust fallback mechanisms must be

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refined to ensure seamless transitions to human operators in cases of system uncertainties or failures.

- **Managing Cost and Infrastructure Requirements**: High-performance, real-time LLM inference and extensive API usage incur significant operational costs that must be balanced against overall system benefits.
- **Data Security and Compliance**: The system must ensure stringent data protection and privacy, adhering to regulations such as GDPR through robust encryption and secure data handling practices.

Further research and iterative testing are planned to optimize each component. Integration of advanced machine learning techniques for error prediction and correction may further enhance system performance and reliability.

VIII. Conclusion and Future Work

A solution for automating taxi phone orders in Bulgaria using an LLM-based conversational AI dispatcher is presented and evaluated. Integration of a dedicated real-time LLM core with robust telephony systems, high-quality speech processing, and seamless external API interfaces results in significant improvements in operational efficiency, reduction of dispatcher workload, and enhancement of the overall customer experience, while maintaining compliance with stringent regulatory and legacy requirements.

Experimental validation in collaboration with Triumph Taxi in Varna confirms that the automated system substantially decreases call durations and improves consistency compared with traditional manual processing. Some performance gaps remain in handling highly complex or ambiguous scenarios, particularly in address recognition accuracy; however, the scalability, cost reductions, and reliability achieved strongly support further development and broader deployment. Ongoing work is directed at closing this gap through expanded data collection from underrepresented dialects and targeted model fine-tuning based on misrecognized cases observed during evaluation.

Future work will focus on refining error handling protocols and enhancing the robustness of the speech and dialogue systems. Additional efforts include expanding the training dataset with more diverse real-world interactions and integrating advanced predictive error correction algorithms to better anticipate and resolve potential failure modes. Continuous system monitoring and user feedback are expected to enable iterative improvements in model performance. Adaptive system tuning based on dynamic operational loads will be explored to ensure optimal responsiveness during peak call periods. Overall, the insights gained are anticipated to contribute to the broader adoption of AI-driven automation in service sectors constrained by legacy processes and regulatory requirements, further streamlining dispatch operations and enabling future advancements in real-time conversational AI applications.

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